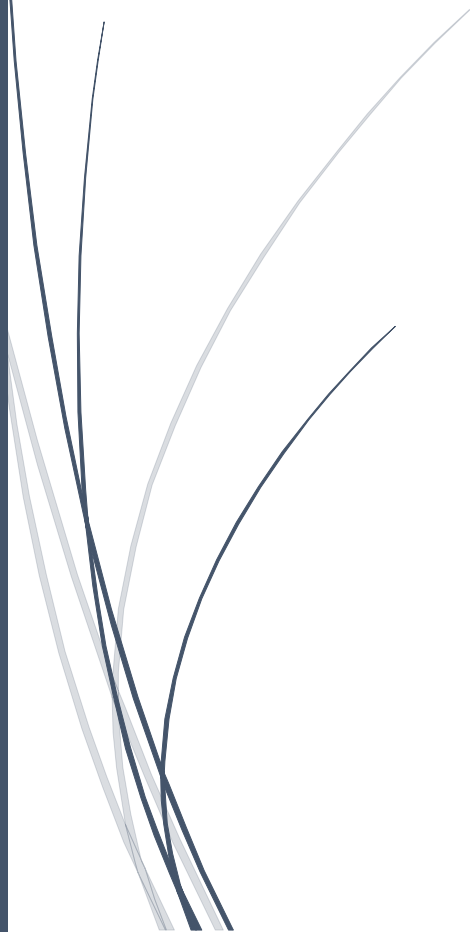


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3/12/2021

Performance-Based Control Co- Design of Building Structures with Controlled Rocking Steel Braced Frames via Neural Dynamic Model

PRJ-2847

Several thin, curved lines in dark blue and light grey originate from the bottom left and sweep upwards and to the right.

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Gutierrez Soto, Mariantonieta

In the optimization model, the following codes and data are provided. In this project the patented Neural Dynamic model (U.S. patent 5,815,394 issued date on September 29, 1998) is employed to obtain the optimal results. According to the patent restriction, the NDAP function which performs the variable updating is not presented. However, one can use any updating strategy in the presented configuration.

- CRSBF-NDAP.py: Initiate and run the optimization procedure
- Obj_Fun.py: Obtain the objective function based on the design variable vector
- Run_Structural_Model.py: Run the structural model and record the time history analysis outputs
- g_constraints.py: Calculate the constraints based on structural simulation outputs
- Element_Mat.txt: Contains the geometrical properties of frame elements including the number, tag, type, and length of elements respectively in columns 1 to 4.
- Section_Mat.txt: Contains the section properties for the W sections used for design. The file includes 10 columns respectively showing the section number, area, major moment of inertia, major radius of gyration, major elastic modulus, minor moment of inertia, minor radius of gyration, minor elastic modulus, web area, and torsional moment of inertia all in SI units.
- Final data: Contains the final optimization data